

STAT 220B — Midterm

Time: 2 hours Open notes

This exam is graded out of 115 points; final scores will be capped at 100. You may use the statement of any earlier part of a question in later parts, even if you did not complete the earlier part.

1. (25 points) Sufficiency and Rao–Blackwell.

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} f(x \mid \theta) = \theta e^{-\theta x}$ for $x > 0$ and $\theta > 0$. We wish to estimate $g(\theta) = 1/\theta$ under squared-error loss $L(\theta, a) = (a - 1/\theta)^2$. Let $T = \sum_{i=1}^n X_i$ and $\bar{X} = T/n$.

- (a) Use the factorization criterion to show that T is sufficient for θ .
State the distribution of T . (5 points)
- (b) Consider $\delta_0(X_1, \dots, X_n) = X_1$. Verify that δ_0 is unbiased for $g(\theta)$ and compute $R(\theta, \delta_0)$. (5 points)
- (c) By symmetry, $\mathbb{E}[X_i \mid T]$ does not depend on i . Use this to compute the Rao–Blackwellized estimator

$$\delta_1(t) = \mathbb{E}[X_1 \mid T = t].$$

(8 points)

- (d) Compute $R(\theta, \delta_1)$ in closed form, and verify that $R(\theta, \delta_1) < R(\theta, \delta_0)$ for all $\theta > 0$ whenever $n \geq 2$. (7 points)

2. (24 points) Bayes rules in the Bernoulli model.

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(\theta)$ with $\theta \in (0, 1)$. Take the prior $\pi(\theta) = \text{Beta}(\alpha, \beta)$ with density $\pi(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$ for $\alpha, \beta > 0$. Let $T = \sum_{i=1}^n X_i$.

- (a) Derive the posterior of θ given $T = t$. Identify it as a named distribution. (6 points)
- (b) Under squared-error loss, derive the Bayes rule $\delta_\pi(t)$. (5 points)
- (c) Consider testing $H_0: \theta \leq 1/2$ versus $H_1: \theta > 1/2$ under the 0–1 loss

$$L(\theta, a_0) = \mathbf{1}\{\theta > 1/2\}, \quad L(\theta, a_1) = \mathbf{1}\{\theta \leq 1/2\}.$$

- (i) Compute the posterior expected loss $\rho(\pi, a_j \mid T = t)$ for each action a_j , $j = 0, 1$. Express your answer in terms of $\mathbb{P}(\theta \leq 1/2 \mid T = t)$. (6 points)
- (ii) Derive the Bayes test: state explicitly the criterion (in terms of $\mathbb{P}(\theta \leq 1/2 \mid T = t)$) for the Bayes action to be a_0 versus a_1 . (7 points)

3. (34 points) Jeffreys prior for the Poisson model.

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$ with $\lambda > 0$, and let $T = \sum_{i=1}^n X_i$. Use squared-error loss $L(\lambda, a) = (a - \lambda)^2$.

- (a) Compute the Fisher information $I(\lambda)$ for this model and write down the Jeffreys prior $\pi_J(\lambda) \propto [I(\lambda)]^{1/2}$. Verify by direct integration that π_J is improper. (6 points)
- (b) Derive the formal posterior $\pi_J(\lambda \mid T = t)$. Identify it as a named distribution and verify that it is proper for every $t \in \{0, 1, 2, \dots\}$. (8 points)
- (c) Derive the generalized Bayes estimator $\delta_J(T)$ under squared-error loss. (7 points)
- (d) Compute $R(\lambda, \delta_J)$ and $R(\lambda, \delta_{\text{MLE}})$, where $\delta_{\text{MLE}}(T) = T/n$. Show that δ_J is uniformly R -dominated by δ_{MLE} , and state the exact size of the gap $R(\lambda, \delta_J) - R(\lambda, \delta_{\text{MLE}})$. What does this say about the admissibility of δ_J ? (13 points)

4. (32 points) Admissibility of a proper Bayes rule.

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$ with $\lambda > 0$, and let $T = \sum_{i=1}^n X_i$. Use squared-error loss. Take the $\text{Gamma}(\alpha, \beta)$ prior with density

$$\pi(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}, \quad \alpha, \beta > 0.$$

- (a) Derive the posterior of λ given $T = t$. Identify it as a named distribution. (6 points)
- (b) Derive the Bayes rule $\delta_{\alpha,\beta}(T)$ under squared-error loss. (5 points)
- (c) Compute $R(\lambda, \delta_{\alpha,\beta})$ in closed form. (9 points)
- (d) Apply Theorem 8 from Lecture 05 (every unique Bayes rule is admissible) to conclude that $\delta_{\alpha,\beta}$ is admissible.
 - (i) Show that the Bayes rule $\delta_{\alpha,\beta}$ is unique.
Hint: under squared-error loss, the posterior expected loss $\rho(\pi, a | t)$ is a strictly convex function of a . What does this imply about its minimizer? (6 points)
 - (ii) Verify the standing assumption $r(\pi, \delta_{\alpha,\beta}) < \infty$ for every $\alpha, \beta > 0$, using the closed form from part (c). (6 points)

Best of luck.